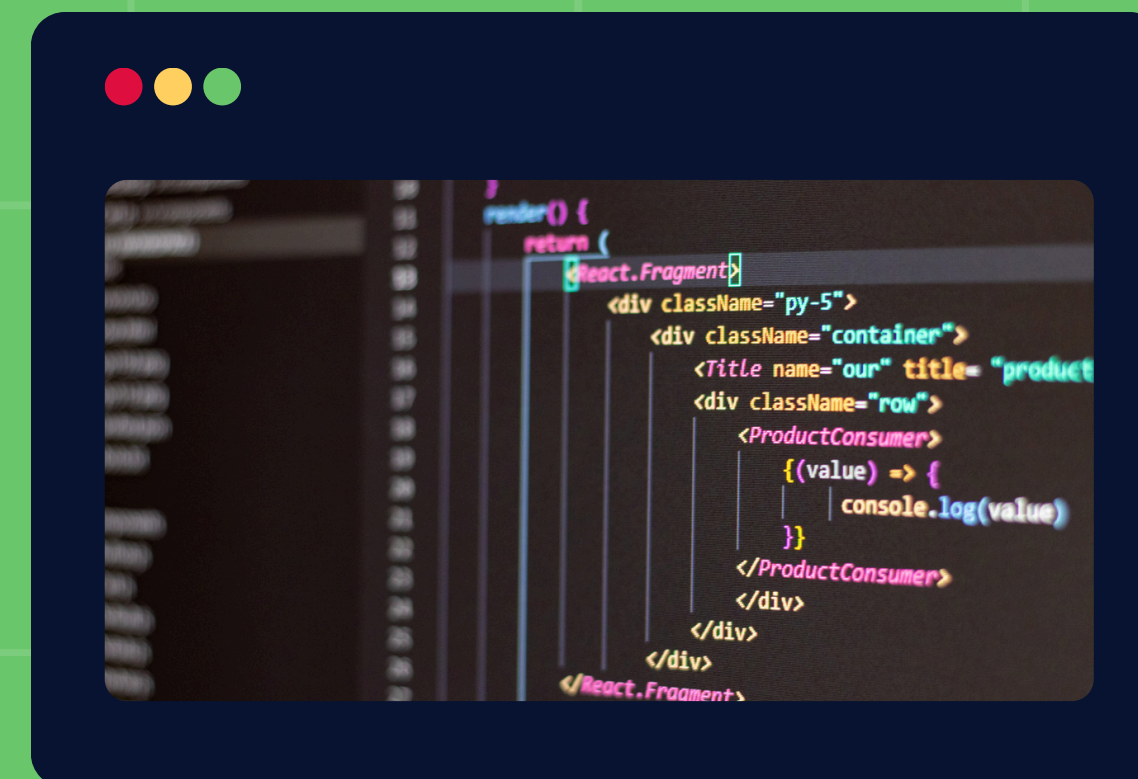




Introduction to Python



April 28th, 2026

IAQCat Formation

Python

Definition

Easy to interpret and high-level object-oriented programming language with easy-to-read syntax

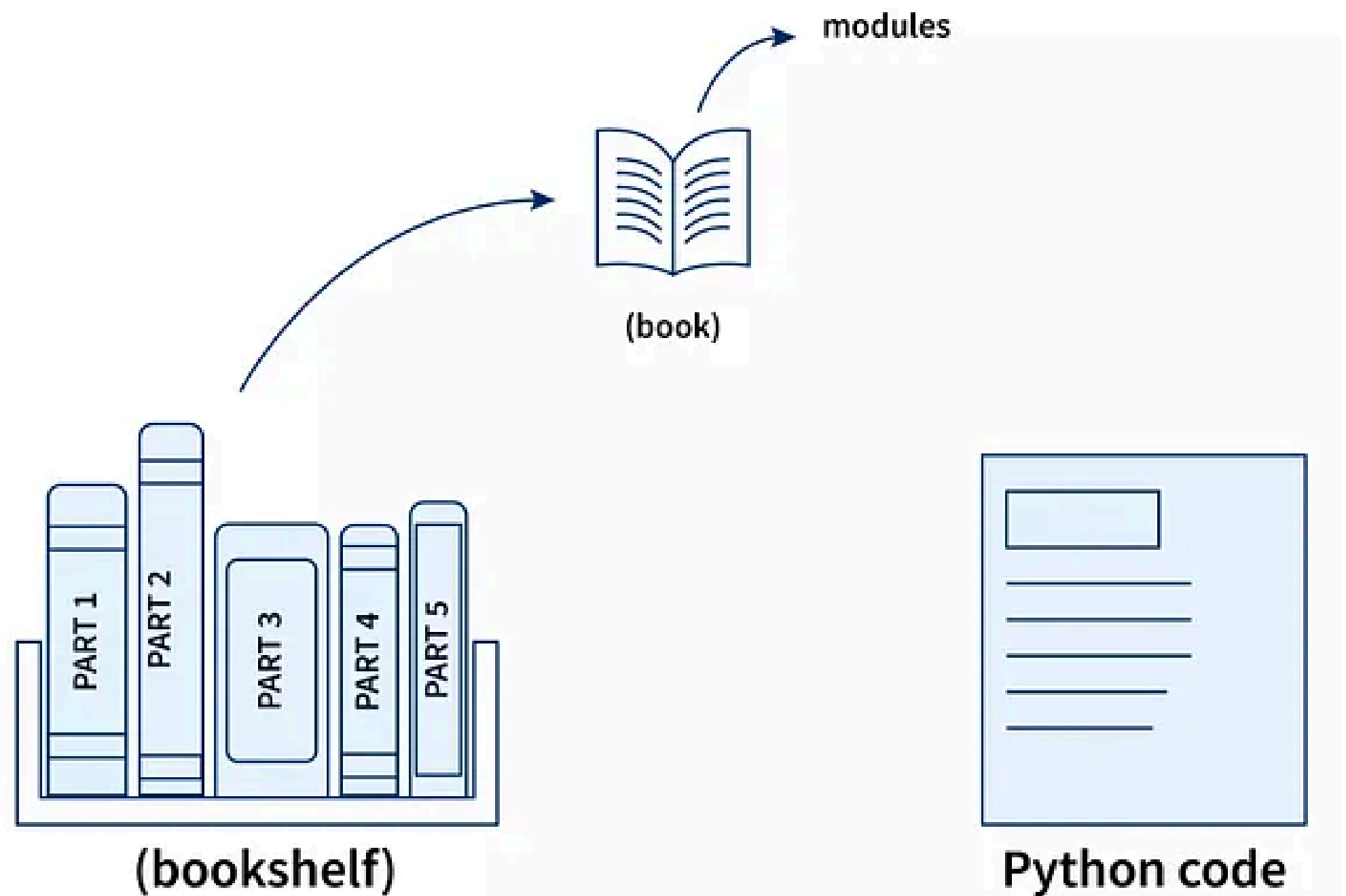
Why Python?

- Simplicity
- Easy interface
- Big toolkit available (modules/packages)



What is a module?

A module is a python file that contains code that define functions, classes and variables.



Common scientific packages



Basics

- NumPy (array data)
- Pandas (tabular data)
- Matplotlib (data visualization)
- Seaborn (data visualization)



Chemistry

- RDKit (cheminformatics and ML)
- Open Babel (conversion of chemical file formats)
- PyMOL (molecular graphics tool)
- ChemPy (physical chemistry)



Machine Learning

- TensorFlow
- Scikit-learn

Virtual environments



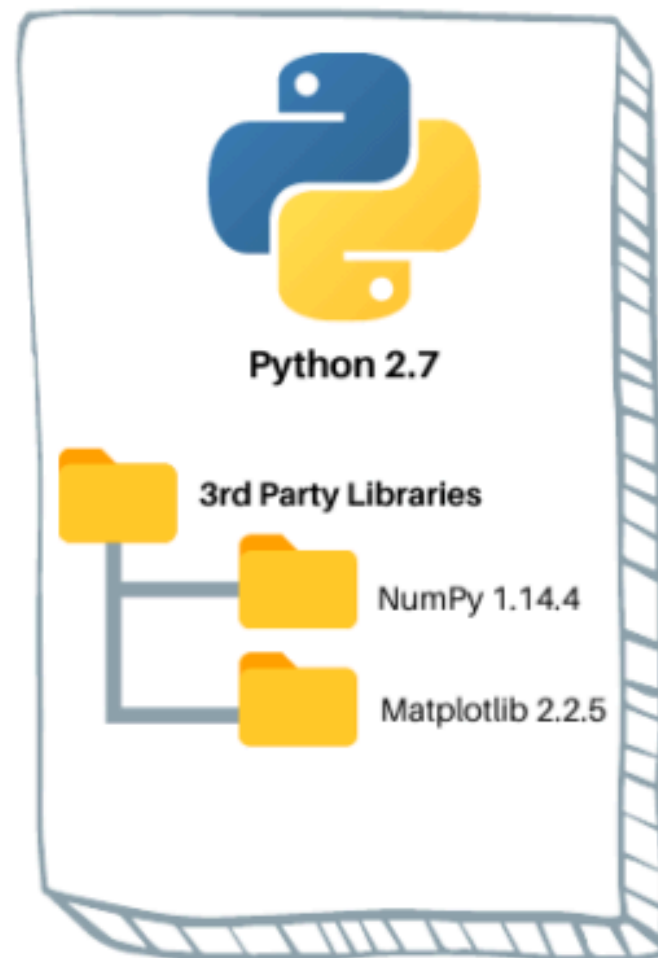
A virtual environment is an isolated directory tree that contains a specific Python version and its own libraries/packages.

Mainly used:

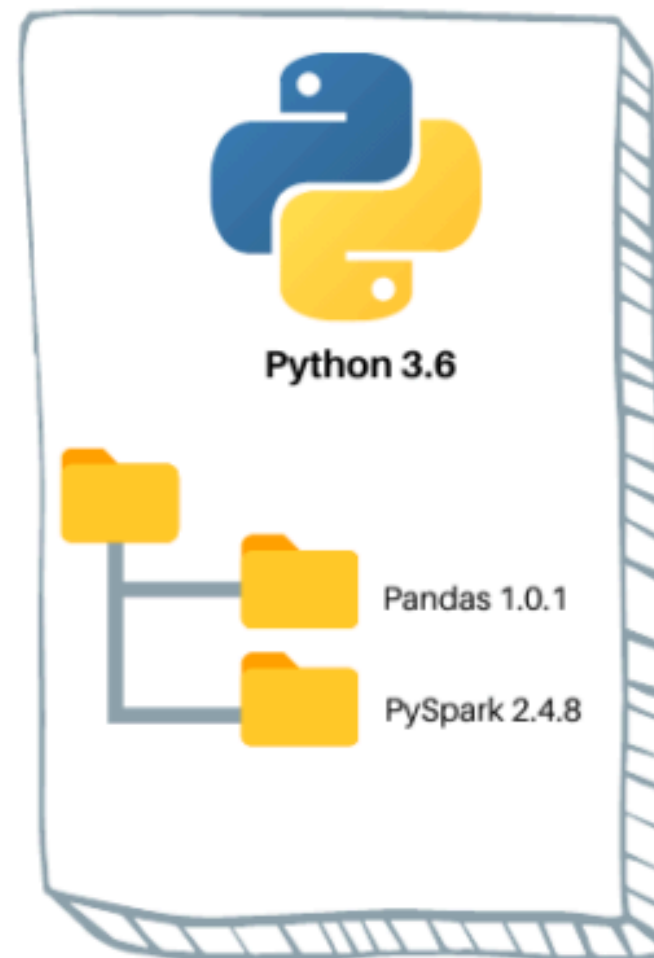
- to avoid dependency conflicts
- for reproducibility



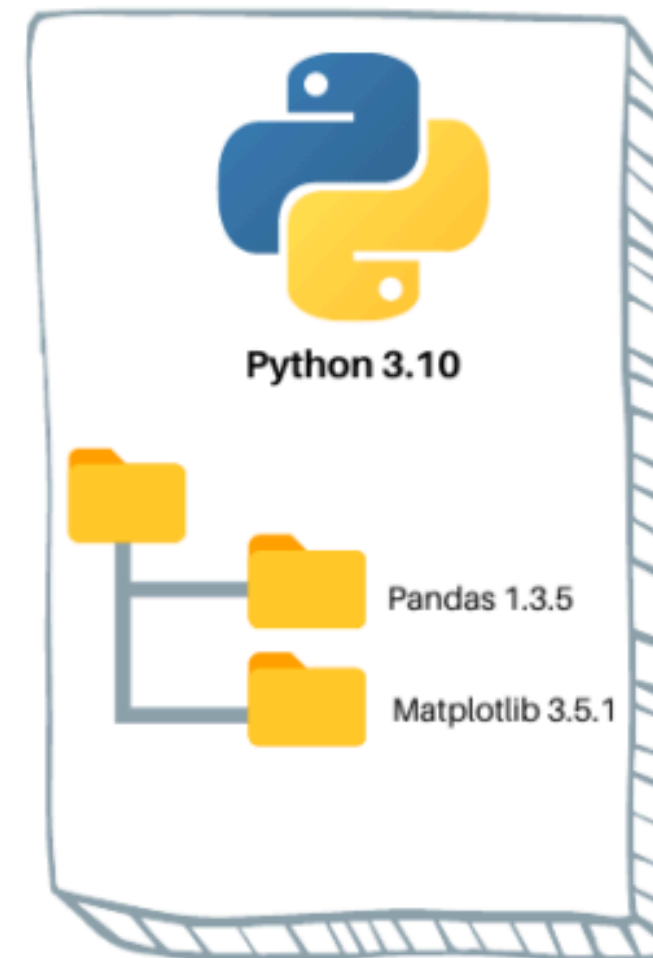
Virtual Environment 1



Virtual Environment 2



Virtual Environment 3



Virtual environment example

Real life example

Personal fumehood

Reagents

Conductions

Lab notebook

Python

Virtual environment

Packages

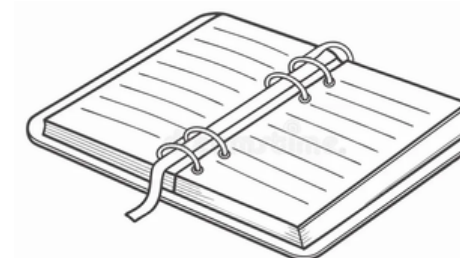
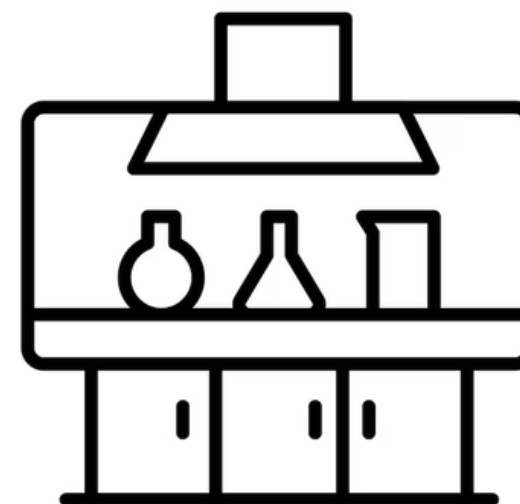
Versions

Requirements (yaml)



Other examples

cheminf
almos



Let's go



Time to open up the introduction to Python jupyter notebook in Visual Studio Code and actually encounter some code!

